

**SECURITIES AND EXCHANGE BOARD OF INDIA
INVESTIGATIONS ENFORCEMENT
& SURVEILLANCE DEPARTMENT
Mittal Court, A Wing, Gr. Floor,
224, Nariman Point, Mumbai 400 021**

**IES/DC/CIR-4/99
28th July 1999**

**To
Executive Directors/ Managing Directors/
Presidents of all Stock Exchanges**

Dear Sir,

SUB : Risk Containment Measures for the Index Futures Market

SEBI, while accepting the recommendations of Dr L C Gupta Committee, had appointed a group under Prof. J R Varma to recommend measures for risk containment for derivatives market in India. The Group focused on ways of making operational the broad recommendations of the Dr. L.C. Gupta Committee to maintain the initial margin to cover 99% Value at Risk. The report provides the methodology for determining initial margin to be charged on Index Futures contracts, prescribes liquid networth, exposure limits for clearing members, transparency and disclosure norms for the clearing corporation and position limits etc. A copy of the Prof. J R Varma Group report is enclosed.

The Board in its meeting held on 19th March 1999 accepted the report of the Prof. J R Varma Group and approved the risk containment measures for the stock index futures market recommended by the Group. The risk containment measures for other derivatives contracts would be prescribed from time to time. The following risk containment measures would have to be complied with and implemented by the derivatives exchange / derivatives segment of an exchange and the clearing corporation / clearing house of an exchange for the index futures trading and settlement :

RISK CONTAINMENT MEASURES FOR THE INDEX FUTURES MARKET

1. **Initial Margin Computation** : The initial margin would be computed based on 99% Value at Risk (VAR). The exponential moving average method would be used to obtain the volatility estimate every day. The estimate at the end of **day t** (σ_t) is estimated using the previous volatility estimate i.e. as at the end of **t-1 day** (σ_{t-1}), and the return (r_t) observed in the futures market during **day t**. The formula would be as under :

$$(\sigma_t)^2 = \lambda (\sigma_{t-1})^2 + (1 - \lambda) (r_t)^2$$

where

- i. λ is a parameter which determines how rapidly volatility estimates changes. The value of λ is fixed at 0.94.
- ii. σ (sigma) means the standard deviation of daily returns in the index futures market.
- iii. The margins for 99% VAR should be based on three sigma limits (three times the standard deviation). The "return" is defined as the logarithmic return : $r_t = \ln(I_t/I_{t-1})$ where I_t is the index futures price at **time t**. The plus/minus three sigma limits for a 99% VAR based on logarithmic returns would have to be converted into percentage price changes by reversing the logarithmic transformation. The percentage margin on short positions would be equal to $100(\exp(3\sigma_t)-1)$ and the percentage margin on long positions would be equal to $100(1-\exp(-3\sigma_t))$. This implies slightly larger margins on short positions than on long positions. The derivatives exchange / clearing corporation may apply the higher margin on both the buy and sell side.
- iv. On the first day of index futures trading the formula given above would require a value of σ_{t-1} , i.e. the estimated volatility at the end of the day preceding the first day of index futures trading. This would be obtained as follows :
 - a. Calculate the standard deviation of returns in the cash index during the last one year.

- b. Set the volatility estimate at the beginning of that year equal to this average value.
 - c. Move forward through the year, one day at a time, using the formula in above to get the estimated volatility at the end of that day using cash index prices.
 - d. The estimated volatility by this method at the end of the day preceding the first day of index futures trading would be the value of σ_{t-1} to be used in the formula given above at the end of the first day of futures trading. Thereafter each day's estimate σ_t becomes the σ_{t-1} for the next day.
- v. For the first six months of index futures trading a parallel estimation of volatility would be done using the cash index prices and the index futures prices and the higher of the two volatility measures would be used to set margins, however, during the first six months, in no case shall the initial margin be less than 5%.
 - vi. The volatility estimated at the end of the day's trading would be used in calculating the initial margin calls at the end of the same day. The volatility estimation and margin fixation methodology should be clearly made known to all market participants so that they can compute what the margin would be for any given closing level of the index. Further, the trading software itself should provide this information on a real time basis on the trading workstation screen.

2. **Margins for Calendar Spreads** : A calendar spread is a position at one maturity which is hedged by an offsetting position at a different maturity e.g. a short position in six months contract matched by a long position in nine month contract.

- i. The margin on calendar spreads shall be at a flat rate of 0.5% per month of spread on the far month contract subject to a minimum margin of 1% and a maximum margin of 3% on the far side of the spread with legs upto 1 year apart.
- ii. A calendar spread should be treated as a naked position in the far month contract as the near month contract approaches expiry. This change should be affected in gradual steps over the last few days of trading of the near month contract. Specifically, during the last five days of trading of the near month contract, the following percentages of a calendar spread shall be treated as a naked position in the far month contract :

100% on day of expiry,

80% one day before expiry,

60% two days before expiry,

40% three days before expiry,

20% four days before expiry.

The balance of the spread shall continue to be treated as a spread. This phasing-in will apply both to margining and to the computation of exposure limits.

- iii. The derivatives exchange should explore the possibility that the trading system could incorporate the ability to place a single order to buy or sell spreads without placing two separate orders for the two legs.

3. **Margin Collection and Enforcement** : The mark to market margins need to be collected before start of the next day's trading. If mark to market margins cannot be collected before start of the next days trading, the clearing corporation/house shall collect correspondingly higher initial margin to cover the potential for losses over the time elapsed in the collection of margins. The clearing corporation /clearing house should lay down operational guidelines for collection of margin and standard guidelines for back office accounting at the level of clearing member and trading member to facilitate the detection of non-compliance at each level. The accounting guidelines shall be in conformity with the guidelines, if any, issued by SEBI from time to time.

2. **Liquid Net Worth and Exposure Limits of a Clearing Member** :

- i. The Liquid Net Worth is defined as under :
 - a. total liquid assets deposited with the exchange / clearing corporation / house towards initial margin and capital adequacy, **LESS**

- b. initial margin applicable to the total gross open positions at any given point of time on all trades to be cleared through the clearing member.
- ii. The clearing member's liquid net worth must satisfy both the Conditions 1 and 2 on a real time basis:
 - a. **Condition 1:** Liquid Net Worth shall not be less than Rs 50 lacs at any point of time.
 - b. **Condition 2:** The mark to market value of gross open positions at any point of time of all trades cleared through the clearing member shall not exceed $33^{1/3}$ times his liquid network.
- iii. Exposure limit for calendar spreads : The Calendar Spread shall be regarded as an open position of one third (1/3rd) of the mark to market value of the far month contract. As the near month contract approaches expiry, the spread shall be treated as a naked position in the far month contract in a phased manner as prescribed above at para 2(ii). If the closing out of one leg of a calendar spread causes the members' liquid net worth to fall below the minimum levels specified, his terminal shall be disabled and the clearing corporation / house shall take steps to liquidate sufficient positions to restore the members' liquid net worth to the levels mandated.
- iv. A numerical example on computation of capital adequacy, exposure limits and initial margin requirements is enclosed at **Annexure - A**.
- v. **Liquid Assets :** At least 50% of the total liquid assets shall be in the form of cash equivalents viz. cash, bank guarantee, fixed deposits, T-bills and dated government securities. Liquid Assets for the purposes of initial margins as well as liquid net worth would include cash, fixed deposits, bank guarantees, Treasury bills, government securities or dematerialised securities (with prescribed haircuts) pledged in favour of the exchange / clearing corporation or bank guarantees as defined hereunder.

A. **Bank Guarantees :** the clearing corporation / house would set an exposure limit for each bank taking into account all relevant factors including the following :

- a. The Governing Council or other equivalent body of the clearing corporation / house shall lay down exposure limits either in rupee terms or as percentage of the trade guarantee fund that can be exposed to a single bank directly or indirectly. The total exposure would include guarantees provided by the bank for itself or for others as well as debt or equity securities of the bank which have been deposited by members as liquid assets for margins or net worth requirement.
- b. Not more than 5% of the trade guarantee fund or 1% of the total liquid assets deposited with the clearing house whichever is lower shall be exposed to any single bank which is not rated P1 (or P1+) or equivalent by a RBI recognised credit rating agency and not more than 50% of the trade guarantee fund or 10% of the total liquid assets deposited with the clearing house whichever is lower shall be exposed to all such banks put together.
- c. The exposure limits and any changes thereto shall be promptly communicated to SEBI. The clearing corporation shall also periodically disclose to SEBI its actual exposure to various banks.

A. **Securities :** The Governing Council or other equivalent body of the clearing corporation / house shall approve the list of acceptable securities, the hair-cuts applicable to various classes of securities, and the method of periodic revaluation (marking-to-market). These policies shall be promptly disclosed to SEBI. The clearing corporation is free to adopt more stringent conditions than those described below.

- a. The marking to market of securities shall be carried out at least weekly for all securities.
- b. Debt securities shall be acceptable only if they are investment grade. Haircuts shall be atleast 10% with weekly mark to market.
- c. The total exposure of the clearing corporation to the debt or equity securities of any company shall not exceed 75% of the trade guarantee fund or 15% of the total liquid assets of the clearing corporation / house whichever is lower. Exposure for this purpose means the mark to market value of the securities less the applicable haircuts.
- d. Equity securities shall be in dematerialised form. The acceptable securities shall be the top 100 securities by market capitalisation out of the top 200 securities by market capitalisation and by trading value. This list shall be updated on the basis of the average market capitalisation over the previous six months. When a security is dropped from the list of acceptable securities, existing deposits of that security will continue to be counted for liquid assets for a period of one month. Haircuts on equity shall be at least 15% with weekly mark to market. The clearing corporation may charge a higher haircut on concentrated

portfolios of equity securities deposited by a member.

- e. All securities deposited for liquid assets shall be pledged in favour of the clearing corporation.

5. Position Limits :

- i. **Customer Level** : Instead of prescribing position limits at the client level, a self-disclosure requirement similar to that in the take-over regulations is prescribed :
 - a. Any person or persons acting in concert who together own 15% or more of the open interest shall be required to report this fact to the exchange and failure to do so shall attract a penalty as laid down by the exchange / clearing corporation / SEBI.
 - b. This requirement may not be monitored by the exchange on a real time basis, but if during any investigation or otherwise, any violation is proved, penalties can be levied.
 - c. This would not mean a ban on large open positions but only a disclosure requirement.
- ii. **Trading Member Level** : There shall be a position limit on the near month contract at the trading member level of 15% of the open interest or Rs 100 crore whichever is higher. This will be reviewed after six months of index futures trading.
- iii. **Clearing Member Level** : No separate position limit is prescribed at the level of clearing member. However, the clearing member shall ensure that his own trading position and the positions of each trading member clearing through him is within the limits specified in para 5(ii) above.
- iv. **Market Level** :
 - a. No limits are prescribed at this stage on the total market wide open interest (as a percentage of the underlying market capitalisation).
 - b. This will be reviewed at the end of six months of index futures trading to determine whether position limits are required at this level to guard against situations where a very large open interest leads to attempts to manipulate the underlying market.

- 6. **Reporting and Disclosure** : The derivatives exchange and clearing corporation shall submit quarterly reports to SEBI regarding the functioning of the risk estimation methodology highlighting the specific instances where price moves have been beyond the estimated 99% VAR limits. The clearing corporation / clearing house shall disclose the details of incidences of failures in collection of margin and / or the settlement dues on a quarterly basis. Failure for this purpose means a shortfall for three consecutive trading days of 50% or more of the liquid net worth of the member.

Any proposal for changes in the methodology to compute the initial margin should be filed with SEBI and released to the public for comments along with detailed comparative backtesting results of the proposed methodology and the current methodology. The proposal shall specify the date from which the new methodology will become effective and this effective date shall not be less than three months after the date of filing with SEBI. At any time up to two weeks before the effective date, SEBI may instruct the derivatives exchange and clearing corporation / house not to implement the change, or the derivatives exchange and clearing corporation / house may on its own decide not to implement the change.

The derivatives exchange / segment of the exchange / clearing corporation / clearing house of the exchange may choose to impose more stringent requirements also than those prescribed above.

Yours faithfully,

L. K. SINGHVI
SENIOR EXECUTIVE DIRECTOR

Annexure A

Numerical Example On Computation Of Capital Adequacy And Initial Margin Requirements

1. Beginning of day one

Suppose that the position at the beginning of day one is as follows:

Member's Liquid Assets	Cash equivalent deposits 35,00,000 Securities deposits (net of haircuts) 40,00,000
Member's Open Position	200 contracts long in the 3 month contract
Futures Prices	3 month contracts is Rs. 1,00,000 1 month contract is Rs. 98,000
Initial Margin	5%
Days to expiry	Fifth day before expiry of one month contract

The margin and capital adequacy calculations will be as follows:

- Initial margin = $5\% * 200 * 1,00,000 = 10,00,000$
- Total open position = 2,00,00,000
- Total liquid assets will be treated as 70,00,000 only since at least 50% of total liquid assets must be in cash equivalents (see Para 4(v)).
- Liquid net worth = $70,00,000 - 10,00,000 = 60,00,000$

Both conditions of networth and exposure limit (given at para 4(ii) in the circular) are satisfied as shown below:

Condition 1. $60,00,000 > 50,00,000$

$$2. 60,00,000 * 33^{1/3} = (20,00,00,000) > 2,00,00,000.$$

2. Initiation of spread trade on day one

Suppose that the member does a calendar spread trade by buying 300 contracts of 3 months futures and selling 300 contracts of 1 month futures.

Since the near month contract of the spread is five days to expiry, the member will have the full benefit of spread margining:

- Margin on spread = $1\% * 300 * 1,00,000 = 3,00,000$
- Spread open position $300 * 1,00,000 * 1/3 = 1,00,00,000$

Adding the figures for the earlier long position we get:

- Total open position = $2,00,00,000 + 1,00,00,000 = 3,00,00,000$
- Liquid net worth = $70,00,000 - 10,00,000 - 3,00,000 = 57,00,000$

Both conditions in para 4(ii) of the circular are satisfied as shown below:

Condition 1. $57,00,000 > 50,00,000$

$$2. 57,00,000 * 33^{1/3} = 19,00,00,000 > 300,00,000$$

3. Margin and capital adequacy calculations on day two

Suppose that on day two, the member does not initiate any new trades, but prices move up so that the situation is as follows:

Member's Liquid Assets	Cash equivalent deposits 35,00,000 Securities deposits (net of haircuts) 40,00,000
Member's Open Position	200 contracts long in the 3 month contract

	300 contracts spread position (long in three month contract and short in near month contract)
Futures Prices	3 month contracts is Rs. 1,01,000 1 month contract is Rs. 99,000
Initial Margin	5%
Days to expiry	Fourth day before expiry of one month contract

The margins and exposures for the 200 contract long position would be:

- Open position = $200 * 1,01,000 = 2,02,00,000$
- Initial Margin = $5\% * 200 * 1,01,000 = 10,10,000$

The spread open position for exposure purposes would be 1,41,40,000 as calculated below since the near contract is four days to expiry:

- 20% of far month = $20\% * 300 * 1,01,000 = 60,60,000$

PLUS

- 80% of spread = $80\% * 1/3 * 300 * 1,01,000 = 80,80,000$

The initial margin on spread would be 5,45,000 as shown below:

- 20% of far month = $20\% * 5\% * 300 * 1,01,000 = 3,03,000$
- 80% of spread = $80\% * 1\% * 300 * 1,01,000 = 2,42,000$

The margin, exposure and liquid network of the member would be as follows:

- Total open position = $2,02,00,000 + 1,41,40,000 = 3,43,40,000$
- Total initial margin = $10,10,000 + 5,45,400 = 15,55,400$
- Liquid net worth = $70,00,000 - 15,55,400 = 54,44,600$

Both conditions in para 4(ii) of the circular are satisfied as shown below:

Condition 1. $54,44,600 > 50,00,000$

2. $54,44,600 * 331/3 (18,14,86,667) > 3,43,40,000$